ModelIt!

Teacher's Guide

Why Do Things Fall Down?

The Science of Gravity and Falling

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-PS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: K-PS2-1

Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students plan a simple drop test to compare what happens when a ball is dropped from low versus high.
Disciplinary Core Idea	**PS2.A Forces and Motion** — Pushes and pulls can have different strengths and directions. Gravity is a pull that acts on all objects near Earth, pulling them downward.
Crosscutting Concept	**Cause and Effect** — Students observe that changing the drop height (cause) changes how fast the ball falls (effect).

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Gravity is one of the most important forces in the universe. It keeps our feet on the ground, makes rain fall from clouds, and holds the Moon in orbit around Earth.

What Is Gravity?

Gravity is a force — an invisible pull — that attracts objects toward each other. On Earth, gravity pulls everything toward the center of the planet. This is why when you jump, you come back down. When you drop a ball, it falls to the floor. Gravity is always pulling on us, even though we cannot see it.

Why Things Fall at Different Speeds

When an object is held up high and released, gravity pulls it downward. The higher the starting point, the longer gravity has to pull on the object, so it reaches a faster speed before hitting the ground. In a vacuum (with no air), all objects fall at the same rate regardless of weight. In air, lighter objects like feathers fall more slowly because air resistance pushes against them.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Drop Height as the external component (we choose how high to hold it) and Fall Speed as the internal component (it changes on its own based on height).

E **Establish Relationships**
Students discover that higher Drop Height leads to faster Fall Speed — a positive relationship.

V **Visualize & Model**
Students build a simple two-part model connecting Drop Height to Fall Speed with a positive arrow.

E **Evaluate Outcomes**
Students test Low Drop vs. High Drop scenarios and observe the speed difference.

R **Revise & Extend**
Students explore what happens when they drop different objects (heavy ball, light ball, feather) from the same height.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever dropped something? What happened? Today we are going to find out WHY things fall down!"*

Watch For

Hold up a ball and drop it. Ask: What just happened? Why did it go DOWN and not UP?

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn about an invisible force called gravity! Gravity is what makes everything fall down."*

Watch For

Introduce the word 'gravity.' Have kids say it together three times. Do a gesture: hands up, then pull down.

Slide 3: Let's Be Scientists!

3 min

Say: *"Scientists ask questions and do experiments to find answers. Our big question is: Why do things ALWAYS fall DOWN?"*

Watch For

Pair-share: Ask a friend what they think. Give 30 seconds, then ask two students to share.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Drop Height is how high we hold the ball — WE get to decide that! Fall Speed is how fast the ball falls — that happens all by itself."*

Watch For

Use sorting mats. Students place Drop Height on the OUTSIDE and Fall Speed on the INSIDE. Act it out: hold ball high, hold ball low.

Slide 5: How Are They Connected?

5 min

Say: *"When Drop Height goes UP, what happens to Fall Speed? Does the ball fall faster or slower? That means it is a POSITIVE connection!"*

Watch For

Demonstrate with a real ball: drop from knee height, then from above your head. Students point thumbs up for positive connection.

Slide 6: What Did We Find Out? 5 min

Say: *"We found out that gravity pulls EVERYTHING down! And when we drop from higher up, it falls FASTER. That is gravity at work!"*

Watch For

Students say: 'When Drop Height goes up, Fall Speed goes up!' Clap it out like a chant. Fill in the sentence frame together.

Slide 7: STEM Challenge 10 min

Say: *"Now YOU get to be engineers! Build a tower and test dropping balls from different levels. Which level makes the biggest bounce?"*

Watch For

Distribute blocks, cups, and soft balls. Groups build towers with 2-3 levels and test drops. Record with smiley faces (small bounce, medium bounce, big bounce).

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Drop Height
- **INTERNAL:** Fall Speed

Relationships

- **Drop Height to Fall Speed:** POSITIVE (+) — When you drop from higher up, the ball falls faster because gravity pulls it for a longer time.

Key Discoveries

- Gravity pulls everything down toward the ground
- When you drop something from higher up, it falls faster
- Gravity never stops — it is always pulling things down

Multiple Choice Pre-Assessment Answer Key

1. What happens when you let go of a ball?

Correct: A) It falls down

Yes! When you let go of a ball, it always falls down to the ground.

2. If you drop a rock, which way does it go?

Correct: C) It goes down

That is right! A rock always falls down when you drop it.

3. What do you think makes things fall to the ground?

Correct: B) Something invisible pulls them down

Good thinking! Something invisible called gravity pulls everything down toward the ground.

4. If you hold a toy up high and let go, what will happen?

Correct: B) It will fall down

Yes! The toy will fall down because gravity pulls it toward the ground.

Multiple Choice Post-Assessment Answer Key

1. What is gravity?

Correct: B) A pull that makes things fall down

That is right! Gravity is a pull that makes things fall down toward the ground.

2. When you drop a ball from HIGH up, what happens?

Correct: B) It falls faster

Yes! Dropping from higher up makes the ball fall faster because gravity has more time to pull on it.

3. In our model, which part do WE get to decide?

Correct: B) Drop Height

Yes! WE choose how high to hold the ball. That is the Drop Height, and it is the outside part of our model.

4. Does gravity ever stop pulling on things?

Correct: C) No, it never stops

That is right! Gravity never stops. It is always pulling on everything, even when things are sitting on a table.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Heavy things fall faster than light things"	In air, heavy and light objects fall at almost the same speed. A bowling ball and a soccer ball dropped together land at nearly the same time. Air resistance (not weight) is what makes feathers float slowly.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Gravity only works when you drop something"	Gravity is always pulling on everything, all the time — even when things are sitting still on a table. The table pushes back up to keep things from falling through.